

Department of Computer Science and Engineering
BRAC University

Set A

Examination: Semester Final
Duration: 1 hour 40 minutes

Semester: Spring 2026
Full Marks: 40

CSE 423: Computer Graphics

Name:	ID:	Section:
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*Answer the following questions. Understanding the questions is part of the examination.
The numbers on the right margin indicate marks.*

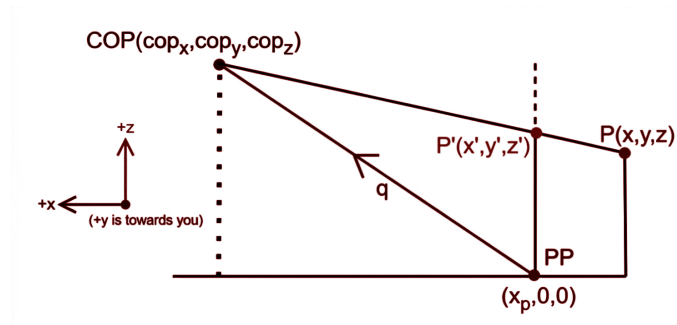
Question 1 [CO3]

- a. While converting from **RGB to HSV**, an intermediate hue value, h , is calculated. Consider an **RGB** color for which the **Cyan** and **Magenta** components (in its equivalent **CMY** representation) are the **largest** and **equal**. **Write down** the formula used to compute h and briefly **explain** how it is derived in no more than three sentences. **4**
- b. A professional digital cinema camera used in filmmaking records video using a **12-bit-per-channel** RGB color system to preserve high dynamic range and smooth color transitions during editing. During the recording of a sunset scene, a pixel is captured with the raw 12-bit RGB values of **(200, 3800, 3800)**. **Convert** this pixel into the **HSV** color space. Based on the computed HSV values and the scene lighting, **determine** how a glossy red sports car would appear in the scene. **5+1**

Question 2 [CO1]

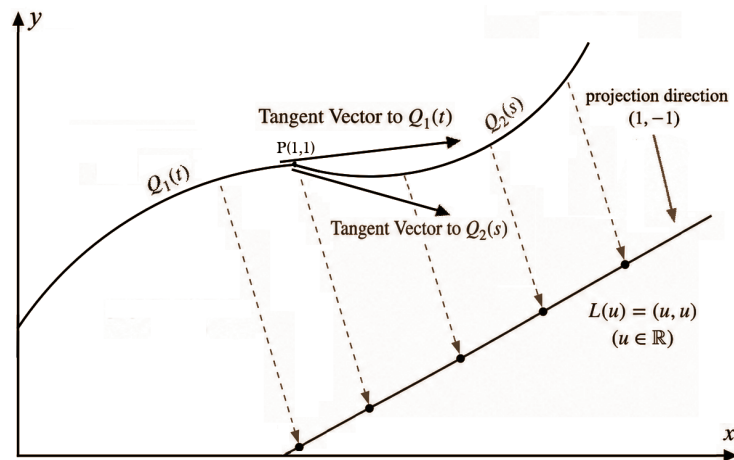
- a. “The **angle of incidence** significantly affects the **intensity of illumination** on a surface.” Briefly **justify** this statement based on **lighting principles** used in computer graphics. **4**
- b. The Professor shines two light sources, **A** and **B**, located at (1, 4, 5) and (2, 5, 5), respectively. The **normal** vector of the plane is given by $\langle 2, 3, 4 \rangle$, and the **point** on the reflective surface under consideration is (-2, 3, 2). Rio is viewing the surface from a direction that produces the **maximum specular highlight** (maximum shininess). He is asked to solve the problem using a variation of the Phong illumination model that involves finding the **bisector (half-vector)** between the light direction vector and the view direction vector. The **ambient**, **diffuse**, and **specular** reflection **coefficients** are **0.3, 0.4, and 0.7**, respectively, and the specular **exponent** is **10**. The **ambient** light in the scene has an **intensity** of **11**. Rio also notices that Light A has an intensity of **7** and illuminates a circular **area** of 115 square units, while Light B has a **diameter** of 10 units and an intensity of **8**. Now, **compute** the **total illumination intensity** I at the surface point **P**, considering only the light source attenuation. **6**

Question 3 [CO2]



- Derive the **4x4 homogeneous matrix** according to the above diagram to obtain a projected point $P'(x', y', z')$ from a point $P(x, y, z)$. 5
- In a perspective projection setup, the **center of projection (COP)** is located at **(18, 42, 15)**, and a point in 3D space is given by **P(520, 30, 70)**. The projection plane is the **YZ plane** placed **300** units from the COP and perpendicular to the viewing direction. Using a **4x4** homogeneous matrix, **determine** the coordinates of the projected point **P'**. 5

Question 4 [CO3]



- Two parametric curve segments joined at a point $P(1, 1)$: 6
 Segment 1: $Q_1(t) = (t, t^2)$; $0 \leq t \leq 1$
 Segment 2: $Q_2(s) = (1 + s, 1 + ks^2)$; $0 \leq s \leq 1$
Find the value of k for which the composite curve is $C(1)$ and $G(1)$ continuous. Briefly **explain** whether the **curvature** will be equal at the joining point or not.
- Suppose the entire curve is projected onto a linear curve, $L(u) = (u, u)$, using parallel projectors in direction $(1, -1)$. **Determine** whether the joint **appears smooth** after projection onto $L(u)$. Briefly **justify** your answer in no more than three sentences. 4